

HILARY OMONDI OTIENO

Backend Engineer & Machine Learning Specialist

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PROFESSIONAL SUMMARY

Backend Engineer & Machine Learning Specialist with hands-on experience building scalable API architectures, data pipelines, and production machine learning models. Proven track record delivering end-to-end ML workflows — from data preprocessing and model tuning to deploying real-time prediction APIs with FastAPI. Proficient in Python, FastAPI, Django REST Framework, PostgreSQL, and core ML frameworks including Scikit-learn and PyTorch. Strong foundation in Linux systems and enterprise network infrastructure, combined with a disciplined, deadline-driven approach to shipping data-driven software solutions.

Core Competencies: Teamwork & Collaboration • Problem-Solving • Discipline & Time Management • Attention to Detail • Adaptability

TECHNICAL SKILLS

- Languages: Python, JavaScript, SQL, HTML/CSS
- ML & Data: Scikit-learn, PyTorch, NumPy, Pandas, Matplotlib — model building, tuning, cross-validation, evaluation
- Backend & APIs: FastAPI, Django, Django REST Framework, Pydantic, REST API design & deployment
- Databases & Tools: PostgreSQL, SQLite, Git, GitHub, Joblib
- Frontend: React
- Systems & Networking: Linux (Ubuntu), Router & Switch Configuration, WAN/LAN Design, SSH Hardening, Wireshark

PROFESSIONAL EXPERIENCE

AnalystLab Africa — Remote Jul 2026 – Present

Machine Learning Intern

- Built and evaluated 6+ classification and regression models (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting) across multiple real-world datasets, achieving up to 82.7% classification accuracy and 0.92 R² on regression tasks.
- Applied hyperparameter tuning (Grid Search, Random Search) and 5-fold cross-validation to systematically compare model performance and improve generalization.
- Deployed a production-ready prediction model via a FastAPI REST API with Pydantic schema validation, serving real-time predictions with sub-second response times.
- Performed end-to-end data preprocessing and EDA — missing value imputation, outlier treatment, feature encoding and scaling — on multiple structured datasets to prepare them for modeling.

Kalanet Communications — Kisumu, Kenya May 2025 – Aug 2025

Industrial Attaché, Network Engineering / IT Operations

- Monitored and maintained network infrastructure, sustaining consistent uptime for enterprise and residential clients.
- Diagnosed and resolved complex connectivity issues under tight response windows, applying structured troubleshooting to isolate root causes quickly.
- Partnered closely with senior network engineers on routing optimization, documenting infrastructure changes for the wider team.

TECHNICAL PROJECTS

Diabetes Risk Prediction API — *FastAPI, Scikit-learn, Pydantic, Joblib*

- Trained and deployed a Random Forest classifier behind a FastAPI REST endpoint, with a Pydantic schema validating all incoming requests and returning predictions with confidence scores in real time.

Student Dropout Predictor — *Python, Scikit-learn, NumPy, Pandas*

- Designed and built a machine learning classification system to identify at-risk university students in Kenya, using academic and financial indicators as predictive features.

NumPy-Net — *Python, NumPy*

- Architected a modular deep learning framework built entirely from scratch using NumPy primitives, implementing forward/backward propagation and custom layers without relying on high-level ML libraries.

Full-Stack E-Commerce Platform — *Django, Django REST Framework, React, JavaScript*

- Built a full-stack e-commerce web application with user authentication, product listings, shopping cart state management, and order processing.

EDUCATION

Bachelor of Science in Computer Science Sept 2022 – Nov 2026 (Expected)

Maseno University, Main Campus

- Relevant Coursework: LAN Design & Implementation, Database Management Systems, Object-Oriented Programming, Data Structures & Algorithms, Software Engineering

Kenya Certificate of Secondary Education (KCSE) Jan 2018 – Apr 2022

Kisumu Boys High School